

FL Simulation Results

File: test_A4

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 75.0% (0.75)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 75.0% (0.75) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8759
Clean + DP Protection Accuracy: 0.8759
Poisoned Accuracy: 0.8080
Poisoned + DP Protection Accuracy: 0.8073
Drop (Clean - Poisoned): 0.0679
Drop (Clean - Init): 0.0831
Drop (Clean DP - Init): 0.0831
Drop (Poisoned - Init): 0.0152
Drop (Poisoned DP - Init): 0.0145
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8816
Clean Recall: 0.8759
Clean F1 Score: 0.8754

Clean DP Precision: 0.8818
Clean DP Recall: 0.8759
Clean DP F1 Score: 0.8754

Poisoned Precision: 0.8179
Poisoned Recall: 0.8080
Poisoned F1 Score: 0.8072

DP Protection Precision: 0.8175
DP Protection Recall: 0.8073
DP Protection F1 Score: 0.8067

Summary

FL Simulation Complete
Task: 17eaa389-c1e4-42d3-910b-ade6c80dd32e
GPU: GPU 0
Init Accuracy: 0.7928
Clean Accuracy: 0.8759
Clean DP Accuracy: 0.8759
Poisoned Accuracy: 0.8080
Data Poison Protection Accuracy: 0.8073
Drop (Clean - Poisoned): 0.0679
Drop (Clean - Init): 0.0831
Drop (Clean DP - Init): 0.0831
Drop (Poisoned - Init): 0.0152
Drop (Poisoned_DP - Init): 0.0145
Data Poison Protection Method: foolsgold
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8816
Clean Recall: 0.8759
Clean F1 Score: 0.8754
Clean DP Precision: 0.8818
Clean DP Recall: 0.8759
Clean DP F1 Score: 0.8754
Poisoned Precision: 0.8179
Poisoned Recall: 0.8080
Poisoned F1 Score: 0.8072
DP Protection Precision: 0.8175
DP Protection Recall: 0.8073
DP Protection F1 Score: 0.8067